41 · Rapid-fire: every answer, back to back

The words of this track, as it says them. Bold brown is the examiner.

This is the rapid-fire track: every answer in this episode, back to back, one breath each. No teaching, no traps, just the answers in the shape you would write them. It is made for the night before, and for the bus on the way in. If one of these sounds unfamiliar, go back to its own track.

Case L, Jack's hearing loss.

The hair cells of the organ of Corti, on the basilar membrane, in the cochlea of the inner ear. Loud, high-amplitude sound bends the hairs against the tectorial membrane until they break off: eighty-five decibels over time, a hundred and thirty at once. No impulses go down nerve eight to the brain, and hair cells don't grow back, so it's permanent. That's sensorineural deafness. The other kind, conduction, is a blockage in the outer or middle ear, like earwax or glue ear, and it's often reversible.

Case L, otosclerosis.

Conduction deafness. The ossicles, the malleus, incus and stapes, are damaged. They sit in the middle ear and carry vibration from the eardrum to the oval window. In otosclerosis they become attached to the walls of the middle ear, so they can't vibrate or amplify the sound, and it isn't conducted to the inner ear. The inner ear and its hair cells are fine, which is the difference from Jack's sensorineural loss.

Case 11, Pania's milk coming in.

Prolactin, from the anterior pituitary, stimulates the alveolar cells to produce milk. Oxytocin, made in the hypothalamus and released from the posterior pituitary, makes the myoepithelial cells contract: the let-down. Both are triggered by the baby suckling. The mammary gland is a modified sweat gland: fifteen to twenty-five lobes around the nipple, lobules of milk-making alveoli, an areola, and suspensory ligaments. Milk goes alveoli, alveolar ducts, lactiferous ducts, nipple. In pregnancy, progesterone builds the lobules and alveoli, and oestrogen grows the ducts.

Case 11, the hormone table.

Ducts and glands grow: oestrogen and progesterone. Blood flow to the breasts: oestrogen. Lobules and alveoli: progesterone. Milk production: prolactin. Milk ejection, the myoepithelial cells contracting: oxytocin. Pipes, factories, fill, taps.

Pedigree charts.

Squares are males, circles are females, shaded is affected. In every generation, never skipping: dominant. Skips a generation: recessive. Males and females about equal: autosomal, and father to son rules out X-linked. Label the homozygous recessives first, then work out everyone else from their parents and children: the affected father is big B little b,

and the mother is little b little b. The Punnett square gives two big B little b to two little b little b: a fifty per cent chance for each child.

Huntington's disease.

A mutated gene on chromosome four, an autosome: C A G repeated thirty-six to a hundred and twenty times, instead of ten to thirty-five. It makes an abnormal, toxic huntingtin protein that builds up in the brain and damages movement, thinking and memory. It's autosomal dominant: one copy is enough, there are no healthy carriers, and it shows in every generation. Big H little h, crossed with little h little h, gives a fifty per cent chance for each child.

Glaucoma and cataracts.

Glaucoma: aqueous humour is made by the ciliary body and drained by the canal of Schlemm. If the drain blocks, fluid builds up, the intraocular pressure rises, and it damages the retina and the optic nerve, which can end in blindness. A cataract: the proteins of the lens denature, so it goes cloudy. Vision becomes blurred and yellowed, with glare. It's treated with surgery that replaces the lens.

The words of genetics.

Homozygous: the alleles are the same. Heterozygous: they're different. Allele: a version of a gene. Dominant: always appears in the phenotype. Recessive: appears only if no other allele is present. Genotype: the alleles inherited. Phenotype: the traits you can see. Autosomal: a disease gene on chromosomes one to twenty-two. X-linked: on the X, and more common in males. Locus: where a gene sits on a chromosome. Chromosomes: lengths of DNA carrying genes, in pairs. Genome: all the DNA in the nucleus. Autosome: a chromosome numbered one to twenty-two. Sex chromosome: the X or the Y. Variation: differences in genes across populations. Inheritance: how genes pass from parents to children.

The Punnett square.

Alleles come in pairs, one from each parent, and the Punnett square shows every possible combination, so the probability of each. One parent across the top, the other down the side. Big B little b crossed with big B little b gives one to two to one: twenty-five per cent homozygous dominant, fifty heterozygous, twenty-five homozygous recessive. So seventy-five per cent brown, twenty-five blue. In codominance, I A and I B are both expressed: I A little i crossed with I B little i gives group A B, group A, group B and group O, twenty-five per cent each.

Patterns of inheritance.

Codominance: both alleles are fully expressed, like A B blood, where I A and I B together give group A B. Incomplete dominance: one allele doesn't fully mask the other, so the heterozygote is in between, like sickle cell carriers with both normal and sickle cells. Polygenic: many genes control one trait, giving a continuous range, like height and skin colour. Mitochondrial: mitochondrial DNA comes only from the mother, to all her children, and severity varies because of the bottleneck effect.

Multiple alleles and X-linked genes.

Multiple alleles: one trait with more than two alleles, like ABO blood: three alleles, six genotypes, four groups. The A and B alleles are codominant and both dominant over O, so AO crossed with BO gives A, B, AB and O, twenty-five per cent each. It isn't polygenic. X-linked: the gene is on the X and the Y has no matching allele, so one copy shows in a male while a female needs two, and only females can be carriers. A carrier mother and a normal father: half the sons affected, and no daughters. An affected father and a normal mother: no affected sons, and every daughter a carrier.

DNA, chromosomes and mutations.

DNA, deoxyribonucleic acid, is a double helix of nucleotides, with adenine paired to thymine and guanine to cytosine. It stores the genetic code and passes traits on. RNA is single-stranded, with ribose and uracil, and carries the code out to make proteins. A chromosome is coiled DNA with many genes, and a gene is a segment that codes for a trait or a protein. A gamete has twenty-three chromosomes and the zygote forty-six, half from each parent. A mutation is a permanent change in a gene's DNA code. A genetic disease comes from a faulty gene, like Huntington's, cystic fibrosis or Tay-Sachs. Aneuploidy is an abnormal number of chromosomes.

Karyotypes and aneuploidy.

A karyotype is a picture of the chromosomes in numbered homologous pairs: forty-six chromosomes, twenty-three pairs, twenty-two pairs of autosomes plus X X or X Y. Aneuploidy is an abnormal number of chromosomes. Down syndrome is trisomy twenty-one, forty-seven chromosomes. Turner syndrome is forty-five, with only one X. It arises from non-disjunction in meiosis: a pair fails to separate, so a gamete gets one too many, or one too few.

Detecting and treating genetic disease.

Genetic testing reads a person's DNA, often from a blood sample: a karyotype, or a CGH array for missing or extra DNA. Chorionic villus sampling takes placenta at eight to twelve weeks. Amniocentesis takes amniotic fluid at fourteen to twenty weeks. Both check the fetal cells. Genetic engineering is the deliberate alteration of DNA, like bacteria altered to make insulin. Gene therapy treats a genetic disease by inserting genetic material into the patient's cells.

The genetics fine print.

A line through a symbol: deceased. A horizontal line: a couple. Their children hang from one line as siblings. Generations are Roman numerals, people numbered left to right. Y-linked: only males, every generation, father to every son. Victoria was a carrier, and Leopold had haemophilia. A trait is a characteristic determined by genes, and a mutagen changes the DNA code. Haploid is twenty-three, diploid forty-six, and pair twenty-three is the sex chromosomes. Mitochondrial DNA is part of the genome, but not the nuclear DNA. Her key: most human traits are determined by multiple alleles.

A one-trait cross is monohybrid. Outside the Punnett square, gametes; inside, zygotes. A test cross uses a homozygous recessive partner, and one recessive offspring proves the parent is heterozygous. Red and white giving pink is incomplete dominance; red and white parts is codominance. Two sickle cell carriers have a twenty-five per cent chance of the lethal form. Polygenic: many genes, each with a small effect. Two non-freckled parents can't have a freckled child. The most common fetal test is amniocentesis; her most reliable is chorionic villus sampling.

The three regions of the ear.

The outer ear: the pinna, the external auditory canal with its cerumen, and the tympanic membrane. It collects sound onto the eardrum and helps locate it. The middle ear: air-filled, in the temporal bone, with the ossicles, which amplify sound onto the oval window. The Eustachian tube equalises pressure, and two small muscles dampen loud sounds. The inner ear: a bony labyrinth of perilymph around a membranous labyrinth of endolymph. The cochlea, with its organ of Corti, hears. The vestibular apparatus balances.

How hearing happens.

The pinna collects sound down the canal, and the tympanic membrane vibrates. The ossicles, malleus, incus and stapes, amplify it onto the oval window. That moves the perilymph, then the endolymph, and the basilar membrane moves, so the hair cells of the organ of Corti bend against the tectorial membrane. That's transduction into nerve impulses, which travel along nerve eight to the auditory cortex in the temporal lobe.

Static and dynamic equilibrium.

Both are sensed by hair cells in the vestibular apparatus of the inner ear, and sent along nerve eight to the brain stem and cerebellum. Static equilibrium is head position relative to gravity, and straight-line movement. It's detected by the maculae in the utricle and saccule, where otoliths slide when the head tilts. Dynamic equilibrium is balance during rotation. It's detected by the crista ampullaris in each semicircular canal, where the endolymph swings the cupula and bends the hairs.

The three layers of the eye.

The fibrous tunic: the sclera, which protects the eye and anchors its muscles, and the cornea, which does about eighty per cent of the refraction. The vascular tunic: the choroid, rich in blood vessels, which nourishes the eye; the ciliary body, which holds the lens, changes its shape, and makes aqueous humour; and the iris, which controls pupil size. The neural tunic, the retina: a pigmented layer that absorbs light, then photoreceptors, rods and cones, then bipolar cells, then ganglion cells, whose axons form the optic nerve.

The retina, and colour blindness.

The photoreceptors are rods and cones, in the retina. Cones sit in the macula lutea, packed in the fovea centralis: bright light, colour, and the sharpest vision. Rods are absent from the fovea and increase towards the edges: dim light, black and white, and many rods share one ganglion cell. Pigments differ by their opsin, and the optic disc is the blind spot. Colour

blindness affects the cones, found mostly in the macula lutea and the fovea centralis: in red-green blindness, the L and M cones.

Near vision.

The ciliary muscle contracts, so the lens bulges and gets rounder: accommodation. The pupils constrict. The eyes converge, turning inwards. For distant vision, all three undo: the ciliary muscle relaxes, the lens flattens, and the pupils dilate. Near light arrives diverging; from over six metres away it's parallel.

Myopia, hyperopia and astigmatism.

Emmetropia: light is focused on the retina. Myopia, short-sighted: distant light focuses in front of the retina because the eyeball is too long, and a concave lens diverges the light. Hyperopia, long-sighted: near light focuses behind the retina because the eyeball is too short, and a convex lens converges the light. Astigmatism: the cornea or lens is curved unevenly, so light focuses at several points and everything is blurred. Glasses, contacts, or laser surgery.

Waves of light and sound.

Amplitude is the wave's height: loudness in decibels, or brightness. Wavelength is peak to peak: the colour of light. Frequency is waves per second, in hertz: the pitch of sound, and a higher frequency means a shorter wavelength. Light is electromagnetic and needs no medium; sound is a mechanical pressure wave that needs one. Reflection: light bounces back, like a mirror. Refraction: light bends as it slows in a different density, like a rainbow, or the cornea and lens focusing light. Sound reflects as an echo or sonar, and refracts, bending away from hot ground.

Taste.

Taste is a chemical sense: taste buds are chemoreceptors for chemicals dissolved in saliva. They sit on the papillae, fungiform, vallate and foliate, and each bud is a cluster of gustatory cells in a supporting capsule, with gustatory hairs in the taste pore. Five tastes: sweet, sour, salty, bitter and umami. The facial nerve carries the front two-thirds, the glossopharyngeal the back third, and the vagus the very back, to the brain stem, the thalamus, the insula and the frontal lobe. Flavour is taste plus smell.

Smell.

The olfactory epithelium is in the roof of the nasal cavity, which is why you sniff. It has olfactory receptor neurons with cilia, supporting cells, and basal cells that replace the neurons. Odourants dissolve in the mucus and bind the receptors. The axons pass through the cribriform plate as cranial nerve one, and synapse in the olfactory bulb, under the frontal lobe. The primary olfactory cortex in the temporal lobe identifies the smell; the amygdala adds emotion, and the hippocampus memory. Total loss is anosmia; partial loss is hyposmia.

The eye, part by part.

The conjunctiva covers the front of the eye and lines the eyelids, and it's infected in pinkeye. The cornea is the clear window where light is first bent: eighty per cent of the refraction, and it can be transplanted. The aqueous humour fills the anterior chamber and drains through the canal of Schlemm. The iris has circular and radial muscle, and blue eyes have less brown pigment. The ciliary body holds the lens, makes aqueous humour, and its involuntary muscles change the lens shape. The lens is biconvex: twenty per cent of the refraction. The vitreous humour is the jelly behind the lens, with the hyaloid canal through it.

The retina is the sensory layer, where the image forms. The macula lutea is the yellow spot, and the fovea, in line with the pupil, has only cones. The optic disc, the blind spot, is where the axons leave as the optic nerve, to the occipital lobe. The choroid is black, to stop reflection, and feeds the retina. The sclera is tough and white, and keeps the eye round. Vision has most of the body's receptors. Strabismus: the eyes can't align. Myopia: too bulgy a lens, biconcave fix. Hyperopia: too flat, biconvex fix.

The ear and sound, fine print.

The pinna is cartilage, with the ear lobe, and channels sound down the external auditory meatus, where ceruminous glands make wax that can cause conduction deafness. The eardrum turns air vibrations into bone vibrations, and the ossicles act as levers; the stapes is the smallest bone in the body. Air in the outer ear, bone in the middle, fluid in the inner. The oval window turns bone vibrations into fluid vibrations, and the round window moves the opposite way. The semicircular canals sit at right angles. The vestibular nerve is balance; the cochlear nerve is hearing.

Perilymph, in the scala vestibuli and tympani, is low in potassium. Endolymph, in the scala media, is high. Potassium and calcium flow into the hair cell, and it releases glutamate. Low pitch is at the apex, high pitch at the base. Sound is mechanical and can't cross a vacuum; light is electromagnetic, made of photons. Light's amplitude is its intensity, in lux; sound's is loudness, in decibels. Red light has the lowest frequency, violet the highest. Refraction bends a wave as part of it slows, like an object in water. Taste and smell are chemoreception, balance is mechanoreception, and taste is eighty per cent smell.

The male reproductive system.

The scrotum keeps the testes three degrees cooler, with the dartos wrinkling the skin and the cremaster lifting the testes. The testes make sperm in the seminiferous tubules and testosterone in the Leydig cells, and the Sertoli cells nourish the sperm. The epididymis stores the sperm and matures them. The vas deferens carries them to the ampulla. The seminal vesicles add sixty per cent of the semen, alkaline, with fructose and prostaglandins, and form the ejaculatory ducts. The prostate, round the urethra under the bladder, adds a third, with citric acid and PSA. The bulbourethral glands clear acidic urine. The urethra carries urine and semen, and the penis, with its corpora cavernosa and corpus spongiosum, delivers the sperm.

Semen, erection and ejaculation.

Semen is sixty per cent from the seminal vesicles, thirty from the prostate, and ten from the sperm and the rest. It carries, feeds, protects and activates the sperm, and it's alkaline. Two to five millilitres holds fifty to a hundred and thirty million sperm per millilitre, and a count below about twenty million is linked to sterility. Erection is parasympathetic: nitric oxide dilates the arterioles, the corpora cavernosa fill, and the veins are squashed shut. Ejaculation is sympathetic: the ducts contract, the bladder sphincter closes, and the bulbospongiosus pumps the semen out. Erectile dysfunction is being unable to attain an erection.

Making sperm and eggs.

Spermatogenesis happens in the seminiferous tubules, continuously from puberty. Spermatogonia divide by mitosis into primary spermatocytes, then meiosis one and two give spermatids, which become four sperm. Oogenesis starts in fetal life: primary oocytes arrest in prophase one until puberty. From puberty, one a month completes meiosis one and is ovulated, and meiosis two finishes only if a sperm enters, giving one ovum. Both are driven by FSH from the anterior pituitary, with LH.

The female reproductive system.

The ovaries hold the follicles in their cortex, make the oocytes, and secrete oestrogen and progesterone. The fimbriae of the fallopian tubes draw in the ovulated oocyte; the tube is the site of fertilisation, and cilia and peristalsis carry the oocyte to the uterus. The uterus, with its fundus, body, isthmus and cervix, receives and nourishes the embryo. Its wall is the perimetrium, the myometrium of smooth muscle, and the endometrium, whose functionalis is shed and whose basalis regrows. The cervix's mucus blocks bacteria. The vagina is the birth canal and the copulation organ, and it's acidic. The vulva is the mons pubis, the labia majora and minora, and the clitoris.

The ovarian and uterine cycles.

The ovarian cycle. In the follicular phase, FSH makes a follicle grow into a Graafian follicle, which secretes oestrogen. At ovulation, around day fourteen, an LH surge ruptures it and releases the secondary oocyte. In the luteal phase the corpus luteum secretes progesterone, and without a pregnancy it becomes the corpus albicans. The uterine cycle. In the menstrual phase, progesterone falls and the stratum functionalis is shed. In the proliferative phase, oestrogen rebuilds it. In the secretory phase, progesterone readies it for implantation. With no implantation, progesterone falls, and menstruation starts again.

The male versus female table.

LH: in females it triggers ovulation; in males the Leydig cells make testosterone. FSH: follicle development, or spermatogenesis via the Sertoli cells. Timing: females start in fetal life and release eggs from puberty to menopause; males go from puberty, for life. Size: the egg is the largest cell, the sperm the smallest. Meiosis: in females it stops and starts and gives one egg; in males it's continuous and gives four sperm. Mobility: the egg is carried by cilia; the sperm swims.

Male hormones and testosterone.

The hypothalamus releases GnRH, so the anterior pituitary releases FSH and LH. LH stimulates the Leydig cells to make testosterone, which enhances spermatogenesis. FSH acts on the Sertoli cells, which nourish the sperm. Testosterone feeds back to switch off GnRH and LH, and inhibin from the Sertoli cells switches off FSH. At puberty, testosterone grows the reproductive organs, and brings pubic, axillary and facial hair, a deeper voice, more bone density and muscle mass, and libido. Females make testosterone too, in the adrenal glands, and it drives libido in both sexes.

Oestrogen, progesterone and the pill.

Oestrogen rises at puberty and drives follicle growth. It grows the uterine tubes, uterus and vagina; it brings breast growth, fat on the hips and breasts, a wider, lighter pelvis, and axillary and pubic hair; and it rebuilds the endometrium in the proliferative phase. Progesterone, from the corpus luteum, readies the lining for implantation, and in pregnancy develops the lobules and alveoli. The pill, synthetic oestrogen and progestin, stops ovulation, thickens the cervical mucus, and thins the endometrium.

The female sexual response, and four words.

The female sexual response is like the male's: the clitoris and vaginal mucosa engorge with blood, the vestibular glands lubricate, and orgasm brings muscle tension, a faster pulse, higher blood pressure, and rhythmic contractions of the uterus. Menarche is the first period. Menopause is when the reproductive hormones decline and the cycles stop. Amenorrhoea is no periods. Dysmenorrhoea is painful periods.

Sperm, eggs and ovulation, fine print.

The sperm head holds twenty-three chromosomes as chromatin, capped by the acrosome, whose enzymes penetrate the egg. The midpiece mitochondria turn fructose into A T P for swimming, and the tail propels it. Sperm don't carry food in their heads. A diploid spermatogonium divides by mitosis into a diploid primary spermatocyte, then meiosis gives haploid secondary spermatocytes and spermatids, and spermiogenesis makes spermatozoa. Inhibin slows production when the count is high. Oogonia divide by mitosis only before birth. Meiosis one gives a haploid secondary oocyte and a polar body, and meiosis two completes at fertilisation. Polar bodies degenerate.

Spermatogenesis is continuous from puberty to death, with equal cytoplasm, and gives four motile sperm. Oogenesis is monthly, menarche to menopause, unequal, and gives one non-motile ovum. It's mitosis first, then meiosis. At ovulation, the LH peak bursts the Graafian, or tertiary, follicle, and the secondary oocyte is pushed to the fimbriae. FSH is highest in the proliferative phase. Oestrogen peaks just before ovulation and stimulates LH. Progesterone dominates the secretory phase. Her key: LH stimulates progesterone from the thecal cells.

Erection, the breast, and her female words.

A parasympathetic trigger, then nitric oxide is released, the central artery, arterioles and spongy tissue widen, more blood flows in, the corpora cavernosa fill and squash the veins, and last the blood drains back out: two, five, four, three, one. An enlarged prostate, not a testicle, makes it hard to urinate. Prostaglandins drive peristalsis in both tracts. Males secrete testosterone, FSH and LH. The breast sits on the pectoralis major. The alveoli make milk under prolactin, and the lactiferous ducts release it under oxytocin. The areola has sebaceous glands and darkens in pregnancy, and the nipple hardens under sympathetic stimuli.

Oestrogen gives primary breast development at puberty, and progesterone further development in pregnancy. Breast development is a secondary sexual characteristic, and the breast has no strong muscles supporting it. Ovulation releases an egg, fertilisation fuses egg and sperm, menstruation sheds the stratum functionalis, and pregnancy is forty weeks. Abortion is the termination of a pregnancy by removal or expulsion of an embryo or foetus. The uterus is hollow and pear-shaped, the ovary makes oestrogen and oocytes, and the placenta feeds the foetus through the umbilical cord. Oestrogen enlarges the external genitalia.

The closed-question extras.

Down syndrome is trisomy twenty-one, Edwards trisomy eighteen. Turner: a woman with one X, forty-five. Klinefelter: a man with X X Y, forty-seven. The most common fetal test is amniocentesis. In the dark, the pupils dilate: radial muscles, sympathetic. In bright light and near vision they constrict: circular muscles, parasympathetic. The lens is biconvex, the image is real and inverted, half the fibres cross at the optic chiasma, and the corpus callosum combines the two images. The eyelids are the palpebrae, lacrimal secretions are tears, and pinkeye is the conjunctiva.

Blepharitis is inflamed oil glands in the eyelids. Retinitis pigmentosa kills the rods first, so night blindness comes first. Macular degeneration loses central vision, and the dry kind has no effective treatment. Diabetes causes cataracts, glaucoma and retinopathy. Glaucoma is treated with beta blockers, conjunctivitis with chloramphenicol, astigmatism with toric lenses, and retinal detachment with laser.

Ototoxic drugs, like gentamicin, furosemide, cisplatin, high-dose aspirin and quinine, kill hair cells: sensorineural deafness. High frequencies are heard at the base of the cochlea, low ones furthest from the oval window. Loud sound is felt as vibration by mechanoreceptors. The Doppler effect is a change in frequency for a moving observer, like a passing siren.

The egg is fertilised in the fallopian tube. Her key calls the outer layer the zona radiata, and calcium lets the sperm through. Conception is most likely on days ten to seventeen, and ovulation kits detect the LH peak. The corpus luteum makes progesterone, oestrogen and relaxin for three months, then the placenta takes over; without a pregnancy it becomes the corpus albicans. A pregnancy test detects H C G. An embryo becomes a foetus at eight weeks. The mini pill never releases oxytocin. Breast cancer risks: little breastfeeding, early periods, late menopause, family history, and men get it too.

That's every answer. Go and get them.